

# Tehnički dosije

## FishyFinds

Internet softverske arhitekture (ISA) · 2021/22

Dosije 1.0.0 · provereno 9. septembra 2026. · smoke 66/66

Potpuni tehnički zapis marketplace-a za ribolovni turizam: cilj predmeta i projekta, stek, arhitektura, domenska pravila, UI po ulogama sa objašnjenim snimcima, konkurentnost, testovi, skalabilnost i rezime.

Spring Boot 2.5.7 + Vue 2 + PostgreSQL

Lokalno: <http://localhost:8080>

**Natalija Simin**

Guest / Customer

**Laslo Uri**

Bungalow / Boat owner

**David Jandrić**

Instructor / Admin

Računarstvo i automatika

# Sadržaj

---

- 01 Kontrola dokumenta
- 02 1. Cilj predmeta
- 03 2. Cilj projekta
- 04 3. Identifikacija
- 05 4. Obuhvat
- 06 5. Tehnološki stek (sa objašnjenjem)
- 07 6. Arhitektura sistema
- 08 7. Domenski model (detalj aplikacije)
- 09 8. Funkcionalna specifikacija po ulozi
- 10 9. Lestvica ocena (ISA 6 → 10)
- 11 10. Korisnički interfejs (živi snimci)
- 12 11. Bezbednost, mejl i podaci
- 13 12. Konkurentnost, testovi i skalabilnost
- 14 13. Build i pokretanje (provera 2026)
- 15 14. Rezime
- 16 15. Povezani dokumenti

## Kontrola dokumenta

| Polje           | Vrednost                                                                                                   |
|-----------------|------------------------------------------------------------------------------------------------------------|
| Naslov          | Tehnički dosije, FishyFinds (ISA)                                                                          |
| Verzija dosijea | 1.0.0                                                                                                      |
| Softver         | FishyFinds / artefakt com.fishyfinds:isa 0.0.1-SNAPSHOT                                                    |
| Predmet         | Internet softverske arhitekture (ISA), FTN UNS, 2021/22                                                    |
| Sastavljeno     | 9. septembra 2026.                                                                                         |
| Autori          | Natalija Simin (gost / klijent), Laslo Uri (vlasnik bungalova / broda), David Jandrić (instruktor / admin) |
| Institucija     | Fakultet tehničkih nauka, Univerzitet u Novom Sadu, Računarstvo i automatika                               |
| Provera         | scripts/full-smoke.ps1 <b>66/66 PASS</b> ; Maven servisni testovi zeleni (JDK 17)                          |
| Lokalni ulaz    | http://localhost:8080                                                                                      |

### 1. Cilj predmeta

**Internet softverske arhitekture (ISA)** je predmet na FTN-u gde tim od tri studenta projektuje i isporučuje višekorisnički veb sistem sa pravom perzistencijom, autentifikacijom i ocenjivanim nefunkcionalnim temama (konkurentnost, DevOps, skalabilnost). Za odbranu se očekuje:

- zajednički proizvod sa **podelom uloga** (svaki student brani svoju personu),
- REST + SPA (ili ekvivalent) nad relacionom bazom,
- dokumentovano ponašanje pri konkurentnom pristupu,
- automatski testovi i CI,
- kratka argumentacija skalabilnosti uz bar jedan konkretan mehanizam u kodu.

Ovaj dosije je arhivski zapis te odbrane: čemu sistem služi, kako je složen, šta pokriva svaki opseg ocena i kako je izgledao živi UI u septembru 2026.

### 2. Cilj projekta

**FishyFinds** je marketplace za ribolovni turizam. Gosti pregledaju obalske bungalove, brodove i kurseve sa instruktorom. Registrovani klijenti rezervišu i otkazuju termine. Oglašivači (vlasnici bungalova, vlasnici brodova, instruktori) objavljuju ponude, uređuju dostupnost, pokreću popustne „brze akcije“, rezervišu za klijenta na licu mesta i čitaju grafikone zauzetosti / prihoda. Administratori moderiraju registracije, recenzije, žalbe, zahteve za brisanje naloga, penale zbog nedolaska, loyalty lestvicu, direktorijum korisnika/ponuda i sistemski udeo prihoda.

Proizvod je namerno jedan Spring Boot proces koji služi Vue 2 SPA iz `static/`, uz PostgreSQL. To je kursni sistem, ne produkcion SaaS: SMTP može biti isključen, lozinke su demo, šema se na svakom startu gradi iz seed-a (`create-drop + data-postgres.sql`).

### 3. Identifikacija

| Stavka               | Detalj                                                                     |
|----------------------|----------------------------------------------------------------------------|
| Prikazni naziv       | <b>FISHYFINDS</b> (crveni znak ribe, crni wordmark)                        |
| Engleski naziv       | FishyFinds — coastal stays & fishing days                                  |
| Srpski naziv         | FishyFinds — obalski smeštaj i ribolovni dani                              |
| Vrsta                | Višekorisnička veb aplikacija (SPA + REST)                                 |
| Backend              | Java 11 target, Spring Boot <b>2.5.7</b> , radi na JDK 17+                 |
| Frontend             | Vue 2 (CDN), Vue Router history, Axios, SweetAlert2                        |
| Mape / grafikoni     | OpenLayers 6, Chart.js                                                     |
| Persistencija        | Spring Data JPA, Hibernate, PostgreSQL baza fishyfinds_db                  |
| Auth                 | Spring Security + JWT (TokenUtils), uloge preko @PreAuthorize              |
| Mejl                 | Spring Mail (verifikacija, rezervacije, admin odluke; greške nisu fatalne) |
| Build                | Maven Wrapper; GitHub Actions; SonarCloud                                  |
| Testovi              | JUnit 5 + Mockito; PowerShell živi smoke (scripts/full-smoke.ps1)          |
| Jezik UI             | Engleski chrome (kursni SPA)                                               |
| Jezici dokumentacije | Srpski latinica (ovaj fajl) · English ( <a href="#">DOSSIER_EN.md</a> )    |

| Autor          | Uloge na predmetu                | Glavne površine                                                                                                                       |
|----------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Natalija Simin | Gost, klijent                    | Katalozi, registracija/aktivacija, pretraga i rezervacija, otkazivanje (≥3 dana), istorija, feedback, praćenje, žalbe, penali, profil |
| Laslo Uri      | Vlasnik bungalova, vlasnik broda | CRUD ponuda, termini, brze akcije, rezervacija za klijenta, mape, kalendar nedostupnosti, /owner-reports                              |
| David Jandrić  | Instruktor, admin                | Kursevi, izveštaji o posetama, loyalty, admin redovi (registracije, recenzije, žalbe, brisanja, penali, direktorijum, prihod)         |

Seed lozinka za sve demo naloge: **password**.

| Email                  | Uloga                   |
|------------------------|-------------------------|
| mail@mail.com          | Klijent                 |
| zokaMagic@mail.com     | Vlasnik bungalova       |
| zokiSumi@mail.com      | Vlasnik broda           |
| vesnaVuki@mail.com     | Instruktor              |
| admin@admin.com        | Admin                   |
| pending.owner@mail.com | Registracija na čekanju |

## 4. Obuhvat

U obuhvatu: pregled tri tipa ponuda; JWT prijava; rezervacija/otkazivanje klijenta; životni ciklus ponude vlasnika/instruktora (kreiranje, izmena blokirana dok ima rezervacija, meko brisanje, termini, brze akcije, rezervacija za klijenta); OpenLayers mape; kalendar zauzetosti / nedostupnosti; Chart.js izveštaji; loyalty; kompletan admin hub; dokumenti konkurentnog pristupa; servisni testovi; keš kataloga za PoC skalabilnosti.

Van obuhvata / poštene granice: produkciono ojačavanje SMTP-a, multi-tenant hosting, savršeno snimanje svakog ugnježđenog admin modala i potpuni ručni prolaz kroz svaki SweetAlert. Živi smoke pokriva **66** API/UI kritičnih provera po ulogama — to je prag verifikacije za ovaj dosije.

## 5. Tehnološki stek (sa objašnjenjem)

| Sloj                        | Izbor                        | Zašto je ovde                                                                                   |
|-----------------------------|------------------------------|-------------------------------------------------------------------------------------------------|
| Spring Boot 2.5.7           | Kontejner aplikacije         | Stek iz vremena predmeta: ugrađeni Tomcat, auto-config za Web, Security, JPA, Mail              |
| Spring Web / Data REST      | HTTP površina                | Kontroleri u controllers/ izlažu /api/**; SPA je statički                                       |
| Spring Security + JWT       | AuthN/Z                      | Bezstanjski API: login vraća token; filter čita Authorization; method security po ulozima       |
| Spring Data JPA / Hibernate | ORM                          | Mapira hijerarhiju User i podtipove Offer na PostgreSQL; @Version za optimističko zaključavanje |
| PostgreSQL                  | RDBMS                        | Relacioni integritet rezervacija nad terminima; seed preko data-postgres.sql                    |
| Spring Mail                 | Obaveštenja                  | Aktivacija, rezervacija, mejl pretplatnicima; umotan tako da nestali SMTP ne ruši rezervaciju   |
| Vue 2 CDN SPA               | UI                           | Brza isporuka bez Node build pipeline-a; komponente u static/                                   |
| Vue Router (history)        | Klijentske rute              | Lepši URL-ovi; SpaForwardController prosleđuje nepoznate GET-ove na index.html                  |
| Axios                       | HTTP klijent                 | Kači JWT; govori sa /api                                                                        |
| OpenLayers 6                | Mape (ocena 7)               | Mape na detalju ponude iz koordinata Location                                                   |
| Chart.js                    | Analitika (ocena 7)          | Grafikoni vlasnika; admin finansije                                                             |
| Spring Cache @Cacheable     | PoC skalabilnosti (ocena 10) | Catalog findAll za bungalove / brodove / kurseve                                                |
| Maven + GHA + Sonar         | DevOps (ocena 9)             | Build, test, statička analiza                                                                   |

### Raspored izvornog koda

```
src/main/java/com/fishyfinds/isa/  
  controllers/    REST (ponude, korisnici, termini, admin, analitika...)  
  service/        Poslovna pravila (rezervacija, mejl, loyalty...)  
  model/          JPA entiteti + enumi  
  repository/     Spring Data interfejsi  
  security/       JWT filter, TokenUtils  
  config/         Security, keš, SPA forward  
src/main/resources/static/      Vue SPA  
src/main/resources/data-postgres.sql  
docs/concurrent-access/         Ocena 8  
docs/scalability/SCALABILITY_POC.md Ocena 10  
scripts/full-smoke.ps1         Živa verifikacija
```

## 6. Arhitektura sistema

### Arhitektura

Slika 1. Pregledač Vue SPA → Spring Boot (:8080) → kontroleri/servisi → JPA → PostgreSQL. JWT je u localStorage i zaglavlju Authorization. Čitanja kataloga mogu ići preko Spring Cache.

### Slučajevi upotrebe

Slika 2. Podela uloga po trima studentima ISA, plus zajedničke teme ocena 7–10.

### Tok rezervacije

Slika 3. Put rezervacije (klijent ili vlasnik-za-klijenta): kontroler → provere u servisu → optimistički/pesimistički lock → commit → mejl po najboljoj mogućnosti.

U runtime-u je **jedan JVM**. Nema posebnog Node servera. Statički sadržaj i REST dele origin localhost:8080, što pojednostavljuje CORS za kursni demo.

## 7. Domenski model (detalj aplikacije)

**Korisnici.** User implementira UserDetails. Podtipovi: Customer, BungalowOwner, BoatOwner, Instructor, Admin, uz Authority / status registracije. Oglašivači čekaju admin odobrenje; klijenti se aktiviraju mejl tokenom kada SMTP radi.

**Ponude.** Apstraktni Offer sa Location, ImageItem, AdditionalService, Price. Konkretni tipovi: Bungalow, Boat (sa Engine), Course (veza na instruktora). Meko brisanje i zabrana izmene dok postoje aktivne rezervacije štite inventar.

**Zakazivanje.** Term je rezervabilni prozor na ponudi (@Version). Reservation vezuje klijenta (ili null za ReservationType.QUICK) za termin. CancelledReservation beleži otkazivanja. OfferUnavailability blokira opsege na kalendaru (ocena 7).

**Moderacija i loyalty.** Complaint, UserFeedback, AccountDeletionRequest, Penal, LoyaltyProgram, loyalty redovi po ulozi, VisitReport, Subscriber (praćenje → mejl na brzu akciju).

### Ključna pravila u servisima

- Otkazivanje samo ako rezervacija počinje za **≥ 3 dana**.
- createQuickAction pravi brzu rezervaciju i šalje mejl pretplatnicima.
- updateBungalow|Boat|Course/{id} odbijen dok postoje aktivne rezervacije.
- getTermsByOfferId vraća termine čiji je endTime još u budućnosti (seed prozori pomereni na **2027**).
- Nedostajući Penal red za klijenta se automatski kreira.
- Duplikat DELETE /api/deleteUser/{id} uklonjen; brisanje ostaje na AdminDirectoryController.

## 8. Funkcionalna specifikacija po ulozi

**Gost.** Hero početna; pregled bungalova / brodova / kurseva; detalj (opis, cena, ocena, mapa); registracija; prijava.

**Klijent.** Pretraga/sortiranje; rezervacija slobodnog termina; predstojeće / istorija; otkazivanje uz pravilo 3 dana; feedback; praćenje akcija; žalbe; penali; profil.

**Vlasnik bungalova / broda.** Moje liste sa Details / Terms / Edit / Actions / Book for client / Delete; dodavanje termina; brze akcije; nedostupnost na kalendaru; /owner-reports; visit report gde postoji.

**Instruktor.** Isti obrazac na kursevima (/my-courses).

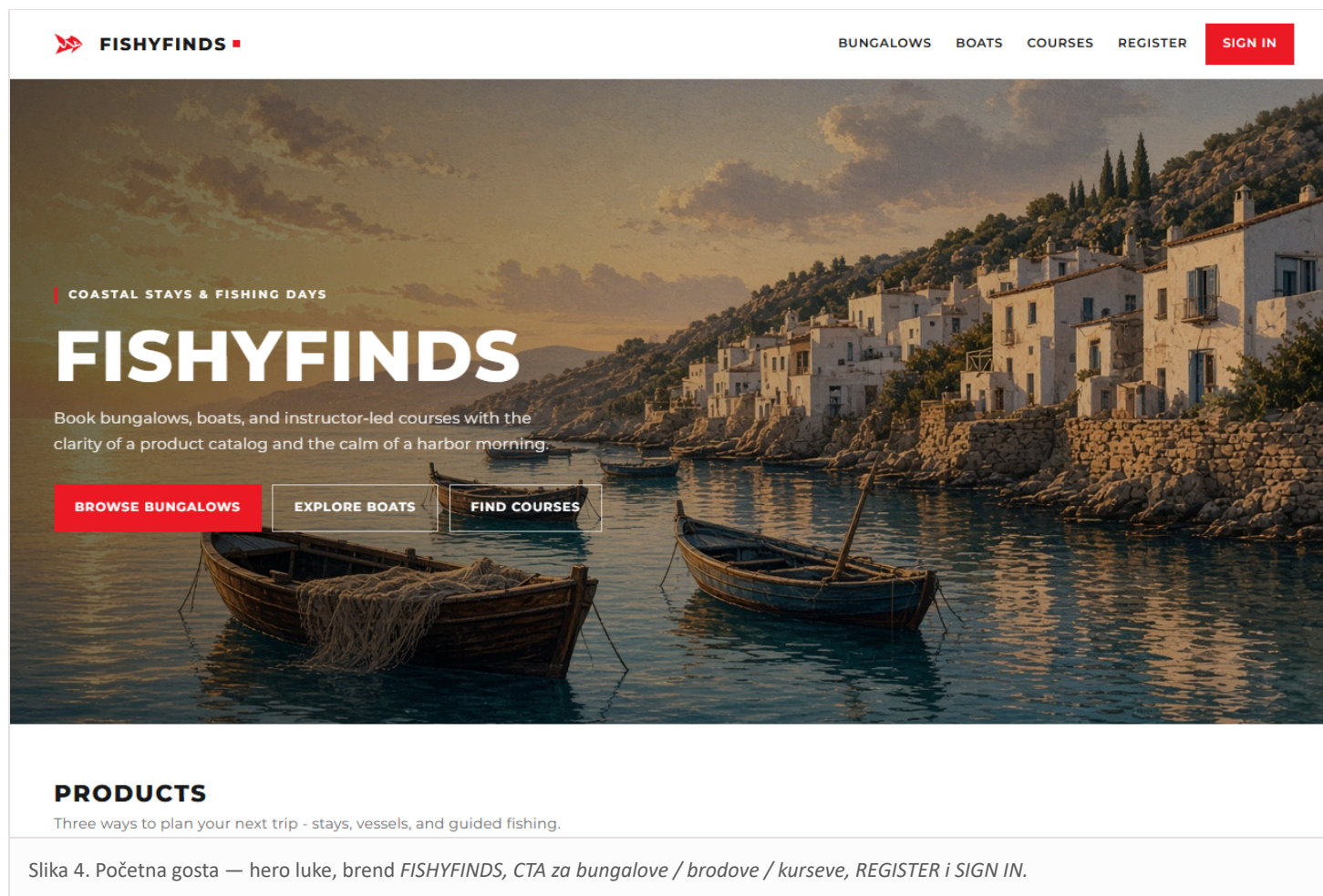
**Admin.** Tile dashboard: registracije, žalbe, recenzije, brisanja, penali, direktorijum, registracija admina, loyalty, prihod / sistemski udeo.

## 9. Lestvica ocena (ISA 6 → 10)

| Ocena | Očekivanje                         | Šta je u ovom stablu                                                                                 |
|-------|------------------------------------|------------------------------------------------------------------------------------------------------|
| 6     | Jezgro marketplace-a               | Terms UI za sve tipove; POST /api/createQuickAction + mejl; PUT update; book-for-client; soft-delete |
| 7     | Mape, grafikoni, kalendar, loyalty | OpenLayers; Chart.js; OfferUnavailability; /admin-loyalty                                            |
| 8     | Konkurentni pristup                | PDF-ovi u docs/concurrent-access/; @Version + lock-ovi                                               |
| 9     | DevOps + testovi                   | GitHub Actions, SonarCloud; Mockito testovi                                                          |
| 10    | Skalabilnost                       | SCALABILITY_POC.md; @Cacheable na catalog findAll                                                    |

## 10. Korisnički interfejs (živi snimci)

Snimci sa lokalne aplikacije 9. septembra 2026. Fajlovi: [../screenshots/](#).



## BUNGALOWS

Browse coastal stays - search, sort, and open details.

| NAME                                                                          | LOCATION                                     | AVAILABLE FROM                              | AVAILABLE UNTIL                             |
|-------------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------|---------------------------------------------|
| <input type="text" value="e.g. Coastal Cabin"/>                               | <input type="text" value="City or country"/> | <input type="text" value="mm/dd/yyyy -:-"/> | <input type="text" value="mm/dd/yyyy -:-"/> |
| SORT RESULTS                                                                  |                                              |                                             |                                             |
| <input type="text" value="Choose order"/>                                     |                                              | <input type="button" value="SEARCH"/>       | <input type="button" value="CLEAR"/>        |
| Leave dates empty to browse all bungalows. Sort updates the list immediately. |                                              |                                             |                                             |



## NIKOS

25 | \* 3.5

Nikos: seaside/mountain bungalow with 1 rooms and 2 beds. Walk to water or trail.

[DETAILS](#)[FOLLOW](#)

## SHREK HOUSE

32.5 | \* 3.6

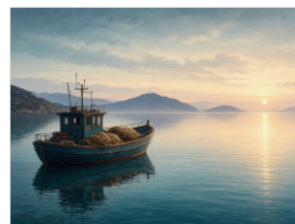
Shrek house: seaside/mountain bungalow with 2 rooms and 3 beds. Walk to water or trail.

[DETAILS](#)[FOLLOW](#)

## FIONA

40 | \* 3.7

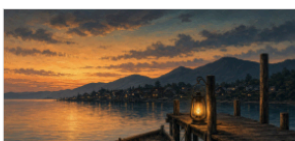
Fiona: seaside/mountain bungalow with 3 rooms and 4 beds. Walk to water or trail.

[DETAILS](#)[FOLLOW](#)

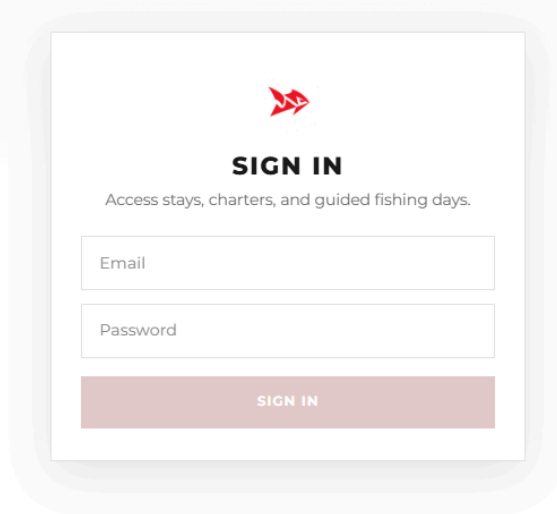
## AZURE CABIN

47.5 | \* 3.8

Azure Cabin: seaside/mountain bungalow with 1 rooms and 5 beds. Walk to water or trail.

[DETAILS](#)[FOLLOW](#)

Slika 5. Javni katalog bungalova — kartice sa fotografijom, cenom, ocenom, lokacijom.



The sign in form is centered on the page. It features the FishyFinds logo at the top, followed by the heading 'SIGN IN' and a subtext 'Access stays, charters, and guided fishing days.' Below this are two input fields: 'Email' and 'Password'. At the bottom of the form is a red 'SIGN IN' button.



## SIGN IN

Access stays, charters, and guided fishing days.

SIGN IN

Slika 6. Prijava — JWT login; demo lozinka password.

## BOATS

Charters and day boats - search, sort, and reserve.

| NAME                                                                      | LOCATION                                     | AVAILABLE FROM                                | AVAILABLE UNTIL                               |
|---------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| <input type="text" value="e.g. Sea Breeze"/>                              | <input type="text" value="City or country"/> | <input type="text" value="mm/dd/yyyy --:--"/> | <input type="text" value="mm/dd/yyyy --:--"/> |
| SORT RESULTS                                                              |                                              |                                               |                                               |
| <input type="text" value="Choose order"/>                                 |                                              | <input type="button" value="SEARCH"/>         | <input type="button" value="CLEAR"/>          |
| Leave dates empty to browse all boats. Sort updates the list immediately. |                                              |                                               |                                               |



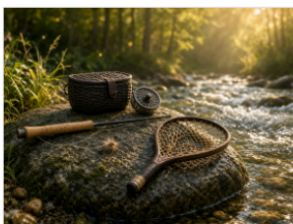
## TITANIK

40 | \* 3.8

Titanik: fishing skiff (5.5m) for coastal fishing and leisure. Engine #1.

DETAILS

FOLLOW



## BLUE MARLIN

52.5 | \* 3.9

Blue Marlin: cabin cruiser (6.2m) for coastal fishing and leisure. Engine #2.

DETAILS

FOLLOW



## SEA BREEZE

65 | \* 4

Sea Breeze: sailboat (6.9m) for coastal fishing and leisure. Engine #3.

DETAILS

FOLLOW



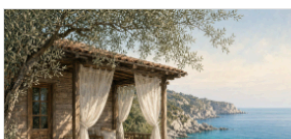
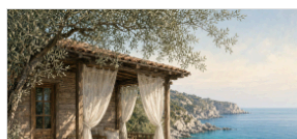
## CAPTAIN'S PRIDE

77.5 | \* 4.1

Captain's Pride: pontoon (7.6m) for coastal fishing and leisure. Engine #4.

DETAILS

FOLLOW



Slika 7. Katalog brodova — isti obrazac kartica.

## COURSES

Instructor-led fishing adventures - browse and book.

| COURSE NAME                                         | LOCATION                                     | INSTRUCTOR                                   | AVAILABLE FROM                                | AVAILABLE UNTIL                               |
|-----------------------------------------------------|----------------------------------------------|----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| <input type="text" value="e.g. Fly fishing intro"/> | <input type="text" value="City or country"/> | <input type="text" value="Instructor name"/> | <input type="text" value="mm/dd/yyyy --:--"/> | <input type="text" value="mm/dd/yyyy --:--"/> |

**SORT RESULTS**

[SEARCH](#) [CLEAR](#)

Leave dates empty to browse all courses. Sort updates the list immediately.



## RACING WITH THE HIPPOS

Vesna Vukelic | 18 | \* 4

Racing with the hippos: instructor-led fishing adventure suitable for 2 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

## DAWN SHORE CASTING

Zeljana Cudesnic | 24 | \* 4.1

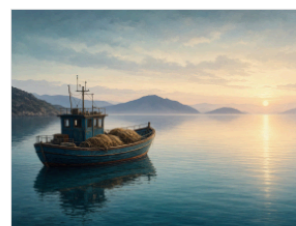
Dawn Shore Casting: instructor-led fishing adventure suitable for 3 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

## KIDS FIRST CATCH

Nemanja Udica | 30 | \* 4.2

Kids First Catch: instructor-led fishing adventure suitable for 4 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

## SPINNING MASTERCLASS

Kristina Mreza | 36 | \* 4.3

Spinning Masterclass: instructor-led fishing adventure suitable for 5 guests. Gear briefing included.

[DETAILS](#)[FOLLOW](#)

Slika 8. Katalog kurseva — kursevi sa instruktorom.

## MAKE A RESERVATION

Search available terms by type, date, duration, and party size.

| OFFER TYPE                                               | START DATE                                                                                                                      | DURATION (DAYS)                     | GUESTS                              |                                       |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|---------------------------------------|
| <input type="text" value="Bungalow"/>                    | <input type="text" value="mm/dd/yyyy --:--"/>  | <input type="text" value="e.g. 3"/> | <input type="text" value="e.g. 2"/> | <input type="button" value="SEARCH"/> |
| Fill all fields to find free terms that match your stay. |                                                                                                                                 |                                     |                                     |                                       |

No available terms yet - adjust filters and search.

Slika 9. UI rezervacije — izbor slobodnog termina i potvrda.

## UPCOMING RESERVATIONS

Your future bookings - search, sort, filter, or cancel.

| OFFER NAME                                                                                                                                                                                                                | SORT RESULTS                                                                                                                                                                                                                              | TYPE                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="text" value="Search bookings"/>                                                                                                                                                                              | <input type="text" value="Choose order"/>                                                                                                                                                                                                 | <input type="text" value="All types"/>                                                                                                                                                                                 | <button>SEARCH</button>                                                                                                                                                                                                                             |
| Sort and type filter apply immediately.                                                                                                                                                                                   |                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                     |
|                                                                                                                                          |                                                                                                                                                          |                                                                                                                                      |                                                                                                                                                                  |
| <b>WAVE DANCER</b><br>2027-09-09T10:00:00 - 2027-09-11T10:00:00<br>Wave Dancer: fishing skiff (9.7m) for coastal fishing and leisure. Engine #7.<br>People: 1   Total: 390   * 4.4<br><button>CANCEL RESERVATION</button> | <b>CYPRESS REST 4</b><br>2026-10-05T10:00:00 - 2026-10-08T10:00:00<br>Cypress Rest 4: seaside/mountain bungalow with 2 rooms and 2 beds. Walk to water or trail.<br>People: 2   Total: 520   * 3.9<br><button>CANCEL RESERVATION</button> | <b>SEA BREEZE 3</b><br>2027-01-09T10:00:00 - 2027-01-12T10:00:00<br>Sea Breeze 3: sailboat (11.1m) for coastal fishing and leisure. Engine #39.<br>People: 2   Total: 840   * 4<br><button>CANCEL RESERVATION</button> | <b>ADVANCED JIGGING</b><br>2027-04-15T10:00:00 - 2027-04-18T10:00:00<br>Advanced Jigging: instructor-led fishing adventure suitable for 4 guests. Gear briefing included.<br>People: 2   Total: 1160   * 4.2<br><button>CANCEL RESERVATION</button> |

Slika 10. Predstojeće rezervacije klijenta — otkazivanje uz pravilo  $\geq 3$  dana.

 FISHYFINDS

MY BUNGALOWS   CALENDAR   REPORTS   VISIT REPORT   MY ACCOUNT   SIGN OUT

### MY BUNGALOWS

Search, sort, and manage your coastal bungalow listings.

BUNGALOW NAME

Search by name

LOCATION

City or country

SORT RESULTS

Choose order

SEARCH

ADD BUNGALOW



**NIKOS**

25 \$ | \* 3.5 | Leptokaria

Nikos: seaside/mountain bungalow with 1 rooms and 2 beds. Walk to water or trail.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



**PINE HAVEN 2**

40 \$ | \* 4.2 | Rovinj 2

Pine Haven 2: seaside/mountain bungalow with 2 rooms and 4 beds. Walk to water or trail.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



**SEAGLASS COTTAGE 3**

55 \$ | \* 4.9 | Igoumenitsa 4

Seaglass Cottage 3: seaside/mountain bungalow with 3 rooms and 2 beds. Walk to water or...

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



**SUNSET LOFT 4**

70 \$ | \* 4.1 | Dubrovnik

Sunset Loft 4: seaside/mountain bungalow with 1 rooms and 4 beds. Walk to water or trail.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



Slika 11. Moji bungalovi — pretraga, ADD BUNGALOW, i dugmad Details / Terms / Edit / Actions / Book for client / Delete.

## MY BOATS

Search, sort, and manage your boat listings.

| BOAT NAME                                   | LOCATION                                     | SORT RESULTS                              |                                                                               |
|---------------------------------------------|----------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------|
| <input type="text" value="Search by name"/> | <input type="text" value="City or country"/> | <input type="text" value="Choose order"/> | <input type="button" value="SEARCH"/> <input type="button" value="ADD BOAT"/> |



**TITANIK**  
40 \$ | \* 3.8 | Zlatibor  
Type: fishing skiff | Length: 5.5 m | Engines: 2 | Power: 100 | Max speed: 35  
Titanik: fishing skiff (5.5m) for coastal fishing and leisure. Engine #1.



**SILVER HOOK 2**  
115 \$ | \* 4.8 | Volos 2  
Type: speedboat | Length: 6.9 m | Engines: 3 | Power: 540 | Max speed: 145  
Silver Hook 2: speedboat (6.9m) for coastal fishing and leisure. Engine #23.



**PELICAN SCOUT 3**  
90 \$ | \* 4.6 | Kragujevac 4  
Type: sailboat | Length: 8.3 m | Engines: 1 | Power: 980 | Max speed: 255  
Pelican Scout 3: sailboat (8.3m) for coastal fishing and leisure. Engine #45.



**LUCKY NET 4**  
65 \$ | \* 4.4 | Pula  
Type: fishing skiff | Length: 9.7 m | Engines: 2 | Power: 1420 | Max speed: 365  
Lucky Net 4: fishing skiff (9.7m) for coastal fishing and leisure. Engine #67.

Slika 12. Moji brodovi — paritet sa upravljanjem bungalovima.

## MY COURSES

Search, sort, and manage the courses you teach.

| COURSE NAME                                 | LOCATION                                     | SORT RESULTS                              |                                                   |
|---------------------------------------------|----------------------------------------------|-------------------------------------------|---------------------------------------------------|
| <input type="text" value="Search by name"/> | <input type="text" value="City or country"/> | <input type="text" value="Choose order"/> | <div><div>SEARCH</div><div>ADD COURSE</div></div> |



**RACING WITH THE HIPPOS**  
18 \$ | \* 4 | Cap 2  
Ohrid, North Macedonia  
Racing with the hippos: instructor-led fishing adventure suitable for 2 guests. Gear briefing included.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



**FLY FISHING INTRO 2**  
42 \$ | \* 4.2 | Cap 4  
Cavtat 2, Croatia  
Fly Fishing Intro 2: Instructor-led fishing adventure suitable for 4 guests. Gear briefing included.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



**REEF BOTTOM FISHING 3**  
66 \$ | \* 4.4 | Cap 6  
Zadar 5, Croatia  
Reef Bottom Fishing 3: instructor-led fishing adventure suitable for 6 guests. Gear briefing included.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



**ADVANCED JIGGING 4**  
36 \$ | \* 4.6 | Cap 3  
Novi Sad, Serbia  
Advanced Jigging 4: instructor-led fishing adventure suitable for 3 guests. Gear briefing included.

DETAILS

TERMS

EDIT

ACTIONS

BOOK FOR CLIENT

DELETE



Slika 13. Moji kursevi (instruktor) — CRUD i isti alati termina/akcija.

## OCCUPANCY CALENDAR

Week strip, month grid, or yearly overview of your bookings and unavailable periods.

### MARK UNAVAILABLE

#### OFFER

Moonlit Lodge 5

#### FROM

mm/dd/yyyy --:--



#### UNTIL

mm/dd/yyyy --:--



SAVE UNAVAILABLE PERIOD

WEEK

MONTH

YEAR

PREV

TODAY

NEXT

September 2026 · Bookings vs unavailable blocks

| MON | TUE | WED | THU | FRI | SAT | SUN |
|-----|-----|-----|-----|-----|-----|-----|
| 31  | 1   | 2   | 3   | 4   | 5   | 6   |

Slika 14. Kalendar vlasnika — zauzetost i blokade OfferUnavailability.

## OWNER REPORTS

Rating, income, and reservation volume at a glance.

AVERAGE RATING

4.3

INCOME

4197 \$

WEEKLY RESERVATIONS

0

MONTHLY RESERVATIONS

0

YEARLY RESERVATIONS

0

1.0

0.9

0.8

0.7

0.6

0.5

Reservation count

Slika 15. Izveštaji vlasnika — Chart.js poslovni grafikoni.


**FISHYFINDS**

MODERATION
FINANCE
MY ACCOUNT
SIGN OUT

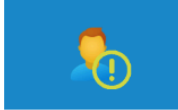
## ADMIN DASHBOARD

Manage registrations, complaints, reviews, penalties, and platform settings.



**REGISTRATIONS**

Approve or deny pending advertiser registrations



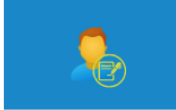
**COMPLAINTS**

Review and resolve customer complaints



**REVIEWS**

Accept or decline pending feedback



**DELETION REQUESTS**

Handle account deletion requests



**PENALTIES**

Approve or decline no-show penalty reports



**DIRECTORY**

Browse and manage users and offers



**REGISTER ADMIN**

Create another administrator account



**LOYALTY**

Configure loyalty categories and rates



**INCOME**

System cut and income overview

Slika 16. Admin dashboard — tile hub za sve redove moderacije i finansije.

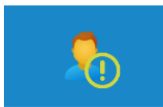
## ADMIN DASHBOARD

Manage registrations, complaints, reviews, penalties, and platform settings.



### REGISTRATIONS

Approve or deny pending advertiser registrations



### COMPLAINTS

Review and resolve customer complaints



### REVIEWS

Accept or decline pending feedback



### DELETION REQUESTS

Handle account deletion requests



### PENALTIES

Approve or decline no-show penalty reports



### DIRECTORY

Browse and manage users and offers



### REGISTER ADMIN

Create another administrator account



### LOYALTY

Configure loyalty categories and rates



### INCOME

System cut and income overview

Slika 17. Red registracija — odobri ili odbij oglašivača.



*Slika 18. Red žalbi — rešavanje uz mogući mejl odluke.*

## SYSTEM INCOME

Platform cut and aggregated reservation income.

CURRENT SYSTEM CUT

5%

NEW SYSTEM CUT (%)

e.g. 10

UPDATE CUT

## INCOME REPORT

TOTAL INCOME

0 \$

SYSTEM INCOME

0 \$

OWNER INCOME

0 \$

RESERVATIONS

0

Slika 19. Prihod / sistemski udeo.

## LOYALTY CATEGORIES

Define point thresholds, discounts, and earning rates.

|                     |                 |                       |              |
|---------------------|-----------------|-----------------------|--------------|
| Category name       | Required points | Category discount (%) | Earning rate |
| <b>ADD CATEGORY</b> |                 |                       |              |

|                                                                                                        |                                                                                                        |                                                                                                           |                                                                                                                 |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>UNBORN SHARK</b><br>Required points: 0<br>Discount: 0%<br>Earning rate: 0<br><b>DELETE CATEGORY</b> | <b>BABY SHARK</b><br>Required points: 50<br>Discount: 20%<br>Earning rate: 0<br><b>DELETE CATEGORY</b> | <b>PARENT SHARK</b><br>Required points: 100<br>Discount: 40%<br>Earning rate: 5<br><b>DELETE CATEGORY</b> | <b>GRANDPARENT SHARK</b><br>Required points: 500<br>Discount: 60%<br>Earning rate: 10<br><b>DELETE CATEGORY</b> |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|

Slika 20. Loyalty program — kategorije i stope (ocena 7).

## 11. Bezbednost, mejl i podaci

- Lozinke sa heširanjem Spring Security; autorizacija JWT + provera uloga.
- Registracija oglašivača ostaje *PENDING* do admin odobrenja.
- Mejl je **best-effort**: uspeh rezervacije ne zavisi od SMTP-a.
- Demo šema: `create-drop + data-postgres.sql`. Termini u seed-u su u **2027.** setval na sekvencama sprečava sudar PK za `loyalty/penal`.

## 12. Konkurentnost, testovi i skalabilnost

**Konkurentnost (ocena 8).** Optimistički `@Version` na `Term / Reservation`; pesimistički `lock` na klijenta gde treba. Opisi: `docs/concurrent-access/`.

**Testovi (ocena 9).** Mockito pokriće servisa; `./mvnw test` na JDK 17 zelen. Živa regresija: `scripts/full-smoke.ps1` (66 provera).

**Skalabilnost (ocena 10).** Pisani PoC (particije, replike, LB, keš). U kodu: `@Cacheable` na `catalog.findAll`.

### 13. Build i pokretanje (provera 2026)

```
# PostgreSQL: baza fishyfinds_db (podrazumevano root/root;  
# override DB_USERNAME / DB_PASSWORD – ne commit-ovati tajne)  
  
./mvnw spring-boot:run  
# otvoriti http://localhost:8080  
  
./mvnw test  
powershell -File scripts/full-smoke.ps1  
powershell -File scripts/capture-screenshots.ps1
```

Preporučen JDK 17+. Aplikacija služi SPA i API na portu **8080**.

### 14. Rezime

FishyFinds je arhiva ISA tima 2021/22: Spring Boot + Vue 2 + PostgreSQL marketplace za bungalove, brodove i ribolovne kurseve, sa jasnom podelom uloga na tri studenta, JWT bezbednošću, admin moderacijom, mapama/grafikonima/kalendarom/loyalty-jem, dokumentacijom konkurentnog pristupa, automatskim testovima/CI i beleškom o skalabilnosti uz keš kataloga. Provereno lokalno 9. septembra 2026. sa 66/66 smoke proverama i zelenim Maven testovima. Ovaj dosije ima englesko i srpsko (latinica) izdanje za usmenu odbranu i kasnije obnove arhive.

### 15. Povezani dokumenti

| Dokument            | Putanja                           |
|---------------------|-----------------------------------|
| Engleski dosije     | <a href="#">DOSSIER_EN.md</a>     |
| Interaktivni HTML   | <a href="#">dossier.html</a>      |
| Konkurentni pristup | ../concurrent-access/             |
| PoC skalabilnosti   | ../scalability/SCALABILITY_POC.md |
| Snimci ekrana       | ../screenshots/                   |
| Dijagrami           | ../assets/diagrams/               |

FishyFinds ISA tehnički dosije — srpsko izdanje.